

Adaptive Modulation Switching in 2D Wireless Communication Channels Under Varying System Parameters

Abstract

Adaptive modulation is a key technique for enhancing wireless communication performance under dynamic channel conditions. This paper presents the design and analysis of an adaptive modulation switching system for a two-dimensional (2D) terrestrial wireless channel using MATLAB simulations.

The proposed system utilizes Signal-to-Noise Ratio (SNR) as the primary metric to dynamically select between Binary Phase Shift Keying (BPSK), Quadrature Phase Shift Keying (QPSK), and 16-Quadrature Amplitude Modulation (16-QAM) through a threshold-based switching mechanism, aiming to balance reliability and spectral efficiency.

The system is evaluated under varying conditions, including distance, transmit power, path loss exponent, carrier frequency, Rician fading (K-factor), and receiver velocity. Results demonstrate that the adaptive scheme maintains reliable communication at low SNR while maximizing data throughput under favorable conditions.

The analysis confirms that adaptive modulation significantly outperforms fixed modulation schemes in dynamic wireless environments and highlights the importance of intelligent modulation selection for efficient and robust communication.

Introduction

Wireless communication systems form the backbone of modern digital connectivity, enabling applications ranging from mobile networks to Internet of Things (IoT) devices. At the core of these systems lies digital modulation, the process of converting binary data into electrical signals suitable for transmission over wireless channels. Common modulation schemes such as Binary Phase Shift Keying (BPSK), Quadrature Phase Shift Keying (QPSK), and Quadrature Amplitude Modulation (16-QAM) provide different trade-offs between reliability and data rate.

BPSK is the most robust modulation scheme, representing one bit per symbol by shifting the carrier phase by 180° , and is highly resistant to noise and interference. QPSK improves spectral efficiency by encoding two bits per symbol using four distinct phase shifts. Higher-order schemes such as 16-QAM further enhance data rates by transmitting four bits per symbol through variations in both amplitude and phase, but at the cost of increased sensitivity to noise.

In conventional wireless systems, a fixed modulation scheme is employed regardless of channel conditions. This leads to a fundamental trade-off between reliability and speed. High-order modulation schemes such as 16-QAM provide higher data rates but suffer from increased Bit Error Rate (BER) under poor channel conditions, especially as the distance between transmitter and receiver increases. On the other hand, lower-order schemes such as BPSK ensure reliable communication but underutilize the available channel capacity when the channel conditions are favorable.

In a 2D terrestrial wireless channel, signal quality is highly dynamic and influenced by multiple factors. One of the primary factors is log-distance path loss, where signal power decreases exponentially with increasing distance. Additionally, environmental effects such as shadowing and multipath fading, caused by obstacles like buildings and terrain, introduce further fluctuations in signal

strength. These variations make fixed modulation schemes inefficient for practical communication systems.

To overcome these limitations, adaptive modulation (AM) has emerged as an effective solution. Adaptive modulation enables the system to dynamically select the most appropriate modulation scheme based on the instantaneous channel condition, typically measured in terms of Signal-to-Noise Ratio (SNR). When the SNR is high, the system utilizes higher-order modulation such as 16-QAM to maximize data rate. As the SNR decreases due to increasing distance or channel degradation, the system switches to QPSK and eventually to BPSK to maintain reliable communication.

This adaptive behavior ensures that the communication system continuously operates at the optimal point between reliability and efficiency. The concept is widely used in modern wireless technologies. For instance, Wi-Fi systems dynamically adjust modulation schemes as users move within different coverage areas, resulting in varying data rates. Similarly, LTE and 5G cellular networks employ Adaptive Modulation and Coding (AMC) techniques to efficiently manage users under diverse channel conditions.

In this paper, an adaptive modulation switching system for a 2D wireless communication channel is developed and analyzed. Unlike conventional studies that focus only on SNR variation, this work investigates the influence of multiple system parameters, including transmit power, path loss exponent, carrier frequency, channel fading characteristics, and user mobility. MATLAB-based simulations are used to evaluate both modulations switching behavior and Bit Error Rate (BER) performance under varying conditions.

The objective of this study is to demonstrate how adaptive modulation improves communication efficiency by intelligently responding to changing channel conditions, thereby achieving an optimal balance between data rate and reliability in practical wireless systems.

System Model

The system model defines the mathematical framework of the 2D terrestrial wireless communication channel used in this study. A Single-Input Single-Output (SISO) communication link is considered, where a transmitter sends data to a receiver moving along a two-dimensional plane. The model captures the effects of large-scale path loss, small-scale fading, and additive noise on the transmitted signal.

Binary data is generated and modulated using appropriate digital modulation schemes. The signal propagates through the wireless channel, where it experiences attenuation, fading, and noise before being received and demodulated. The system performance is evaluated in terms of Bit Error Rate (BER).

Path Loss Model

The primary source of signal attenuation is modeled using the log-distance path loss model:

$$PL(d) = PL_o + 10n \log_{10}(d) \quad (1)$$

where:

- **PL(d)** is the **path loss in dB**
- **PL_o** is the **reference path loss (taken as 30 dB)**
- **n** is the **path loss exponent**
- **d** is the **distance between the transmitter and receiver in meters**

In this study, the path loss exponent n varies between free-space propagation and urban environment to analyze the effect of environmental conditions on system performance.

Signal-to-Noise Ratio (SNR)

The Signal-to-Noise Ratio (SNR) determines the quality of the communication link and is given by:

$$\text{SNR (dB)} = P_t - \text{PL}(d) \quad (2)$$

where:

- P_t represents the **transmit power in dB**
- $\text{PL}(d)$ is the **path loss**

The computed SNR serves as the key parameter for adaptive modulation switching.

Channel Model (Rician Fading)

To represent realistic wireless channel conditions, a Rician fading channel is considered. This model includes both a Line-of-Sight component and multiple scattered components due to reflections and obstacles.

The received signal is expressed as:

$$r(t) = h(t)s(t) + n(t) \quad (3)$$

where:

- $s(t)$ is the **transmitted signal**
- $h(t)$ is the **channel coefficient representing Rician fading**
- $n(t)$ is **Additive White Gaussian Noise (AWGN)**
- $r(t)$ is the **received signal**

The Rician channel is characterized by the K-factor, which represents the ratio of power in the Line-of-Sight component to the scattered components. In this work, **K-factor values of 3, 8, and 12** are used to evaluate the impact of fading conditions on system performance.

Doppler Effect and Mobility

The effect of receiver motion is incorporated through the Doppler shift, which introduces time variation in the channel. The Doppler frequency is given by:

$$f_d = v / \lambda \quad (4)$$

where:

- **v** is the **receiver velocity**
- **λ** is the **wavelength of the carrier signal**

The wavelength is related to the carrier frequency as:

$$\lambda = c / f_c \quad (5)$$

where:

- **c** is the **speed of light** (3×10^8 m/s)
- **f_c** is the **carrier frequency**

An increase in velocity results in higher Doppler shift, causing rapid channel variations and increased BER.

Adaptive Modulation Strategy

The system employs an SNR-based adaptive modulation scheme to dynamically select the appropriate modulation technique. The detailed switching criteria and algorithm are described in Adaptive Modulation Switching Section.

This strategy ensures that lower-order modulation schemes are used under poor channel conditions to maintain reliability, while higher-order schemes are selected in favorable conditions to maximize data throughput.

System Assumptions

The system model is developed under the following assumptions:

- Perfect synchronization between transmitter and receiver
- Ideal channel estimation for accurate SNR calculation
- No interference from other users
- Noise is modeled as Additive White Gaussian Noise (AWGN)

These assumptions simplify the analysis while allowing clear observation of adaptive modulation behavior.

Modulation Schemes Used

In this section, the digital modulation schemes used in the proposed adaptive modulation framework are described. Each modulation technique represents a different trade-off between power efficiency, bandwidth efficiency, and robustness to noise. The selected schemes—Binary Phase Shift Keying (BPSK), Quadrature Phase Shift Keying (QPSK), and 16-Quadrature Amplitude Modulation (16-QAM)—enable dynamic adaptation based on channel conditions.

Binary Phase Shift Keying (BPSK)

Binary Phase Shift Keying (BPSK) is the simplest form of digital modulation, where the carrier signal undergoes a phase shift of 180° to represent binary data. Due to the maximum separation between constellation points, BPSK provides high robustness against noise and interference.

The transmitted BPSK signal is represented as:

$$s(t) = \sqrt{2E_b/T_b} \cos(2\pi f_c t + \theta) \quad (6)$$

where:

- **$s(t)$: Transmitted BPSK signal**
- **E_b : Energy per bit**
- **T_b : Bit duration (time required to transmit one bit)**
- **f_c : Carrier frequency**
- **t : Time variable**
- **θ : Phase of the carrier signal**

The phase θ determines the transmitted bit:

- **$\theta = 0$ represents binary '1'**
- **$\theta = \pi$ represents binary '0'**

The term $\sqrt{2E_b/T_b}$ represents the signal amplitude, which is proportional to the energy per bit. The cosine term represents the carrier wave, while the phase shift encodes the digital information.

This structure ensures maximum separation between the two signal states, making BPSK highly robust against noise and interference.

Each symbol represents **1 bit**, i.e., $\log_2(2) = 1$. The Bit Error Rate (BER) for BPSK in an AWGN channel is given by:

$$\text{BER} = Q(\sqrt{2E_b/N_0}) \quad (7)$$

In the proposed adaptive system, BPSK is used as the fallback mode when **SNR < 10 dB** to ensure reliable communication under poor channel conditions.

Quadrature Phase Shift Keying (QPSK)

Quadrature Phase Shift Keying (QPSK) improves spectral efficiency by encoding two bits per symbol using four different phase shifts of the carrier signal.

The QPSK signal can be expressed as:

$$s(t) = \sqrt{2E_b/T_b} \cos(2\pi f_c t + \theta_k) \quad (8)$$

where:

- $s(t)$: Transmitted QPSK signal
- E_b : Energy per bit
- T_b : Bit duration
- f_c : Carrier frequency
- t : Time variable
- $\theta_k \in \{0, \pi/2, \pi, 3\pi/2\}$ represents the four phase states

Alternatively, QPSK can be represented in in-phase and quadrature form as:

$$s(t) = \sqrt{E_b/T_b} [I \cos(2\pi f_c t) + Q \sin(2\pi f_c t)] \quad (9)$$

where:

- I and Q are the in-phase and quadrature components
- $I, Q \in \{+1, -1\}$

Each symbol represents **2 bits**, i.e., $\log_2(4) = 2$. The BER performance of QPSK in an AWGN channel is identical to BPSK:

$$BER = Q(\sqrt{2E_b/N_0}) \quad (10)$$

In the proposed adaptive system, QPSK is selected when **10 dB $\leq$ SNR < 18 dB**, providing a balance between data rate and reliability under moderate channel conditions.

16-Quadrature Amplitude Modulation (16-QAM)

16-Quadrature Amplitude Modulation (16-QAM) is a higher-order modulation technique that varies both the amplitude and phase of the carrier signal, enabling higher data transmission rates.

The transmitted signal is given by:

$$s(t) = \sqrt{2/T_s} [I \cos(2\pi f_c t) + Q \sin(2\pi f_c t)] \quad (11)$$

where:

- **s(t): Transmitted 16-QAM signal**
- **T_s: Symbol duration**
- **f_c: Carrier frequency**
- **t: Time variable**
- **I and Q are amplitude components**
- **I, Q ∈ {±1, ±3}**

Each symbol represents **4 bits**, i.e., $\log_2(16) = 4$. Due to the closer spacing between constellation points, 16-QAM is more sensitive to noise and channel impairments.

The approximate BER for M-QAM is given by:

$$\text{BER} \approx (4/\log_2 M)(1 - 1/\sqrt{M}) Q(\sqrt{(3 \log_2 M/(M-1)) \cdot (E_b/N_o)}) \quad (12)$$

For 16-QAM (**M = 16**), this simplifies to:

$$\text{BER} \approx (3/4) Q(\sqrt{0.8 E_b/N_o}) \quad (13)$$

In the proposed adaptive system, 16-QAM is used as the high-performance mode and is selected when **SNR ≥ 18 dB**, ensuring high data rates under favorable channel conditions.

Comparative Analysis

The selected modulation schemes exhibit distinct performance characteristics:

- BPSK offers maximum robustness but lowest data rate
- QPSK provides a balance between reliability and spectral efficiency
- 16-QAM achieves high data rates but requires high SNR

This variation enables efficient adaptive switching across different channel conditions.

Summary of Modulation Characteristics

The key characteristics of the modulation schemes used in this work are summarized below:

Parameter	BPSK	QPSK	16-QAM
Bits per Symbol (k)	1	2	4
Constellation Points (M)	2	4	16
Spectral Efficiency	1 bps/Hz	2 bps/Hz	4 bps/Hz
Required SNR	< 10 dB	10–18 dB	> 18 dB
Robustness	High	Moderate	Low

Adaptive Modulation Switching

Switching Logic

The proposed system employs a threshold-based switching mechanism. Based on the computed SNR, the modulation scheme is selected as follows:

BPSK, if $\text{SNR} < 10 \text{ dB}$

QPSK, if $10 \leq \text{SNR} < 18 \text{ dB}$

16-QAM, if $\text{SNR} \geq 18 \text{ dB}$ (14)

At low SNR values, BPSK is selected due to its robustness against noise. For moderate SNR values, QPSK provides a balance between reliability and spectral efficiency. At high SNR values, 16-QAM is used to maximize data throughput.

Working Principle

The adaptive modulation process follows a continuous feedback-based approach:

1. The transmitter sends modulated data through the wireless channel.

2. The receiver estimates the channel quality by measuring the SNR.
3. Based on the estimated SNR, the receiver determines the optimal modulation scheme.
4. This information is fed back to the transmitter.
5. The transmitter switches to the selected modulation scheme for subsequent transmissions.

This process ensures that the system continuously adapts to changing channel conditions.

Algorithm Description

The adaptive modulation algorithm implemented in MATLAB follows a structured procedure:

1. Initialize system parameters such as distance, transmit power, and path loss exponent.
2. Compute path loss using equation (1).
3. Calculate SNR using equation (2).
4. Apply switching logic based on SNR thresholds:
 - a. If $\text{SNR} < 10 \text{ dB}$ → Select BPSK
 - b. Else if $10 \leq \text{SNR} < 18 \text{ dB}$ → Select QPSK
 - c. Else → Select 16-QAM
5. Modulate the data using the selected scheme.
6. Transmit the signal through the channel.
7. Compute BER at the receiver.
8. Repeat the process for different system parameters.

This algorithm enables dynamic adaptation of the modulation scheme in response to varying channel conditions.

MATLAB Implementation

The proposed adaptive modulation logic is directly implemented in MATLAB using conditional statements. The switching mechanism is realized using “if-elseif” conditions based on the computed SNR values.

For example:

- if $\text{snr_dB} < 10 \rightarrow$ modulation index set to BPSK
- else if $\text{snr_dB} < 18 \rightarrow$ modulation index set to QPSK
- Otherwise $\rightarrow$ modulation index set to 16-QAM

This implementation ensures efficient and real-time switching between modulation schemes during simulation.

Results and Analysis

This section presents the results obtained from MATLAB simulations of the proposed adaptive modulation system. The simulation results provide a comprehensive analysis of how environmental and system parameters influence modulation switching behavior and Bit Error Rate (BER) performance in a 2D wireless channel.

Analysis of Modulation Switching Behavior

The “Modulation vs Distance” plots demonstrate the effectiveness of the SNR-based switching logic in adapting to varying channel conditions.

Effect of Distance:

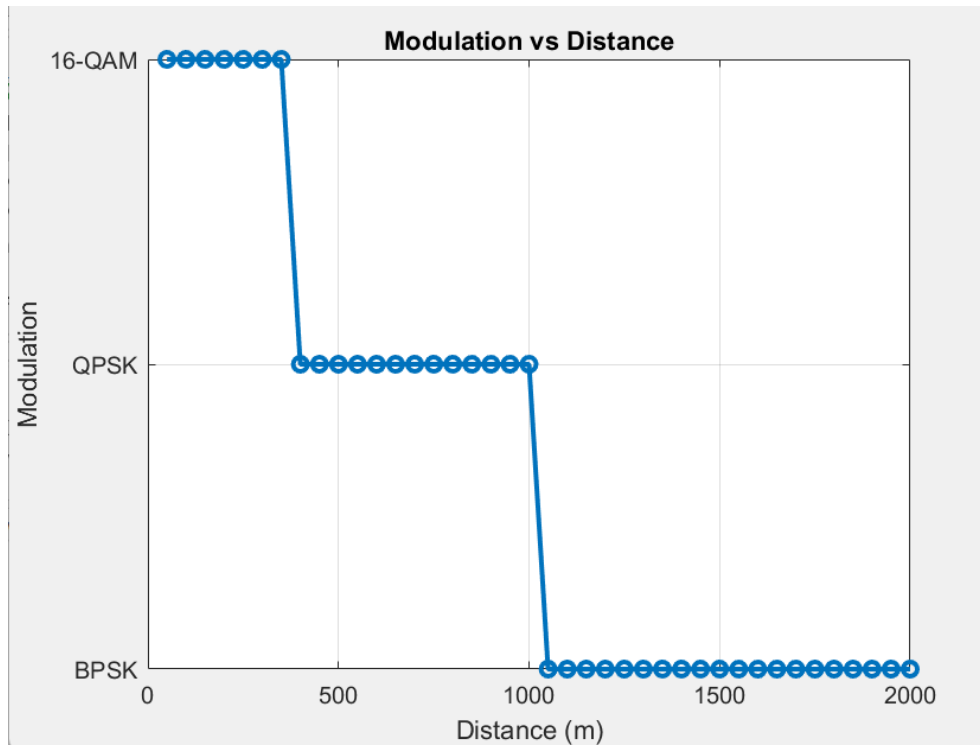


Figure 1: Modulation vs Distance

At shorter distances, the Signal-to-Noise Ratio (SNR) is high due to low path loss, allowing the use of 16-QAM for higher data rates. As distance increases, path loss increases according to equation (1), reducing SNR and forcing the system to switch to QPSK and eventually BPSK. This confirms that the adaptive system maintains reliability at long distances.

Impact of Transmit Power:

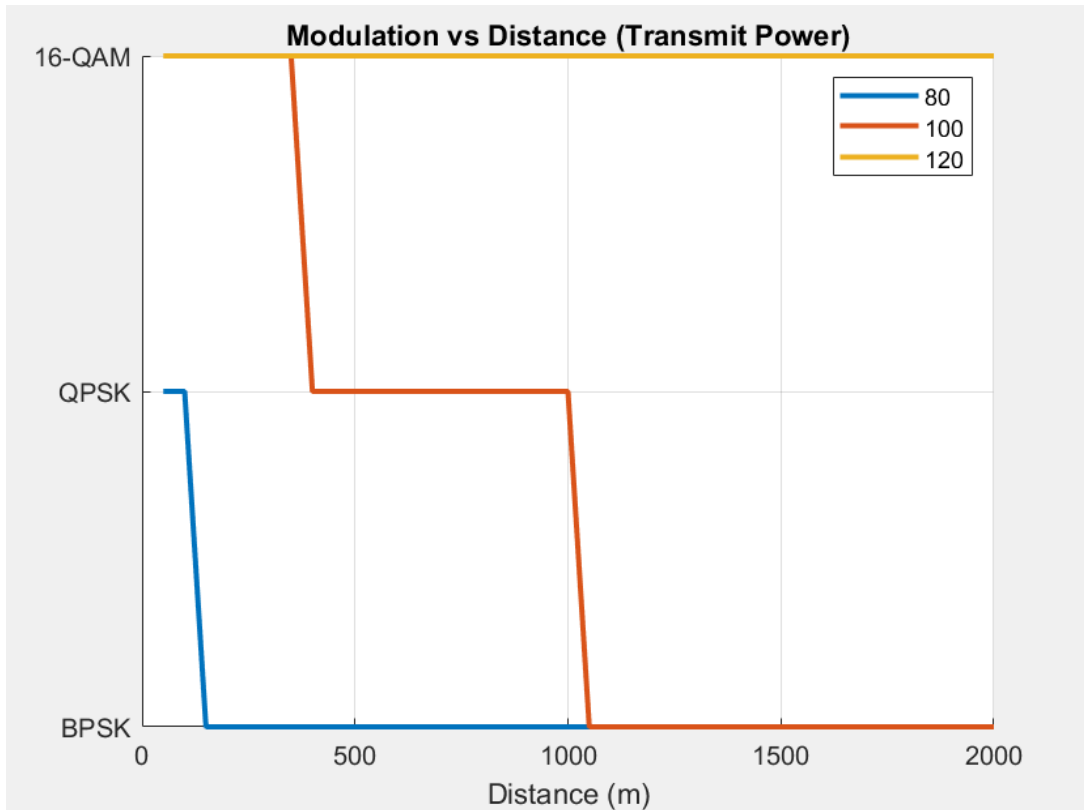


Figure 2: Modulation vs Distance (Transmit Power)

As observed in Figure 2, increasing the transmit power from 80 dB to 120 dB significantly extends the range of high-order modulation. At 120 dB, the system sustains 16-QAM over nearly the entire 2000 m range, whereas at 80 dB, it switches to BPSK within the first few hundred meters. This demonstrates that higher transmit power improves coverage and allows efficient utilization of higher-order modulation schemes.

Effect of Path Loss Exponent (n):

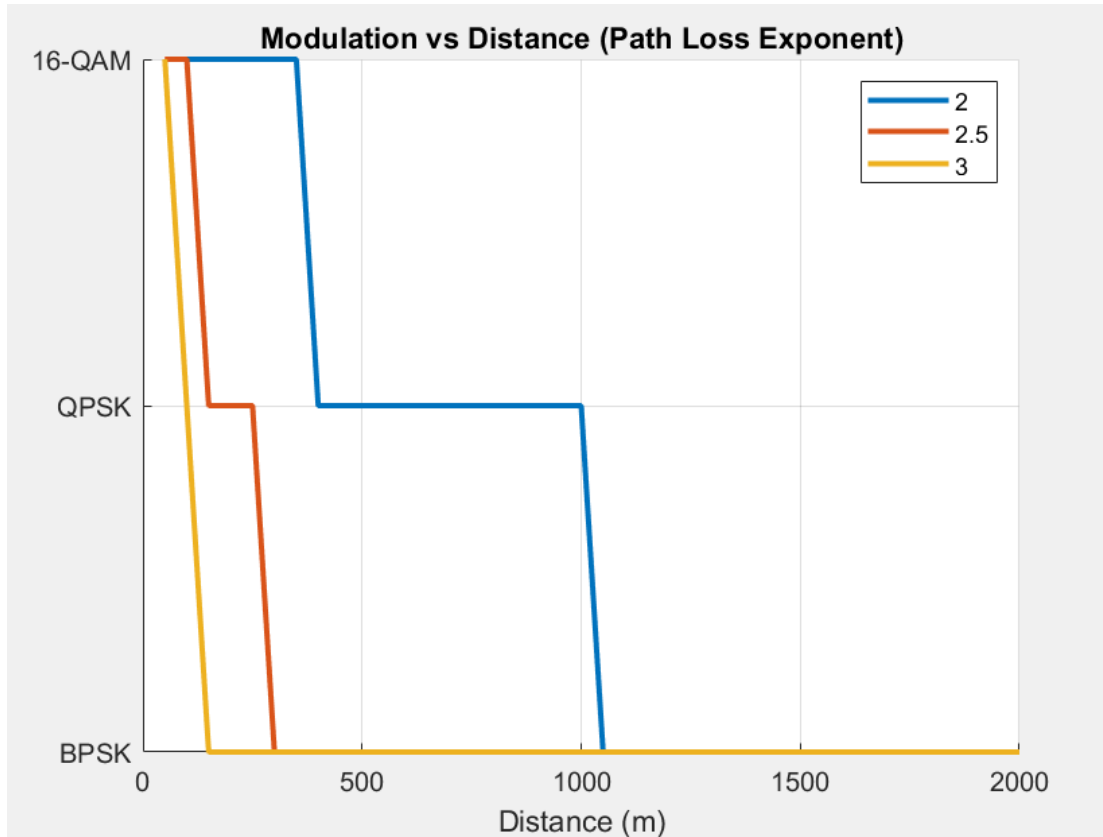


Figure 3: Modulation vs Distance (Path Loss Exponent)

Figure 3 illustrates the impact of environmental conditions. In a free-space environment ($n = 2$), 16-QAM is maintained up to approximately 400 m. However, in an urban environment ($n = 3$), the signal attenuates rapidly, causing the system to switch to BPSK within a much shorter distance (around 100 m). This highlights the sensitivity of adaptive modulation to environmental factors.

Effect of Carrier Frequency:

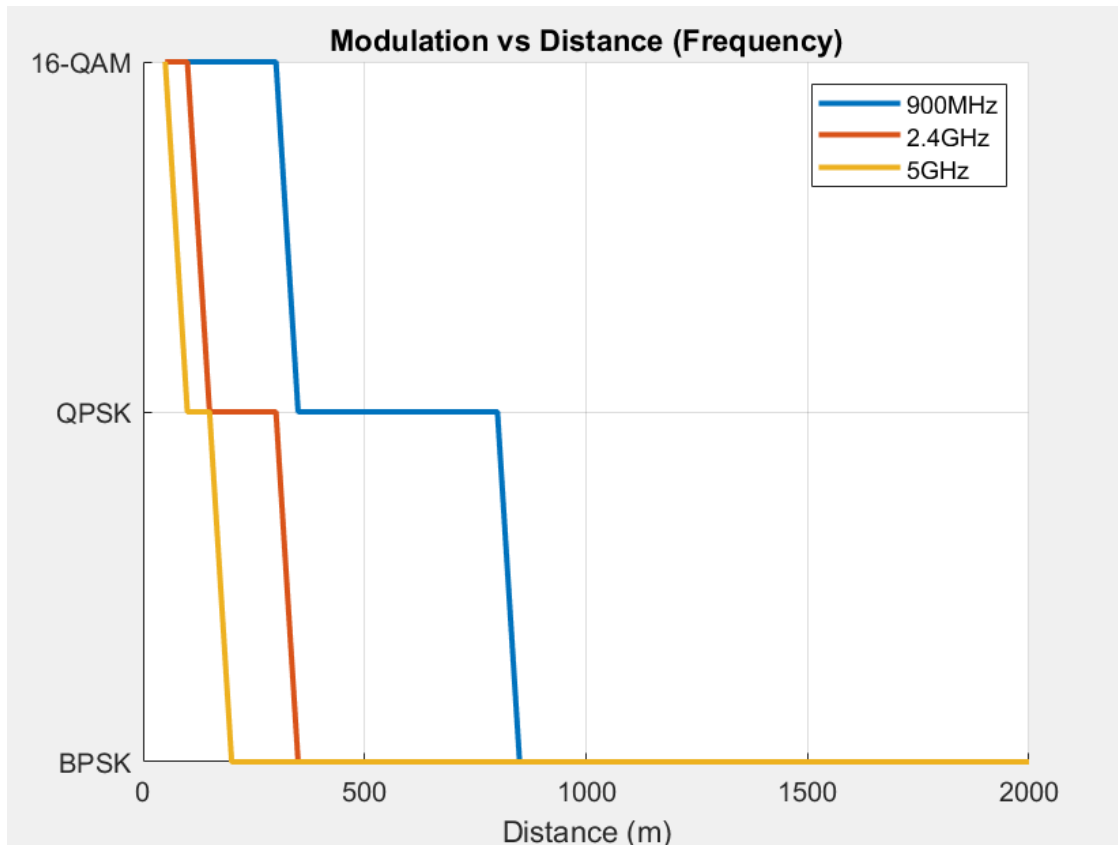


Figure 4: Modulation vs Distance (Frequency)

Figure 4 shows that higher frequencies (5 GHz) experience greater attenuation compared to lower frequencies (900 MHz). As a result, higher frequencies require earlier switching to robust modulation schemes such as BPSK. Lower frequencies allow higher-order modulation to be sustained over longer distances, making them more suitable for long-range communication.

Bit Error Rate (BER) Performance

The BER plots validate the reliability of the system under fading and mobility conditions.

Influence of K-Factor:

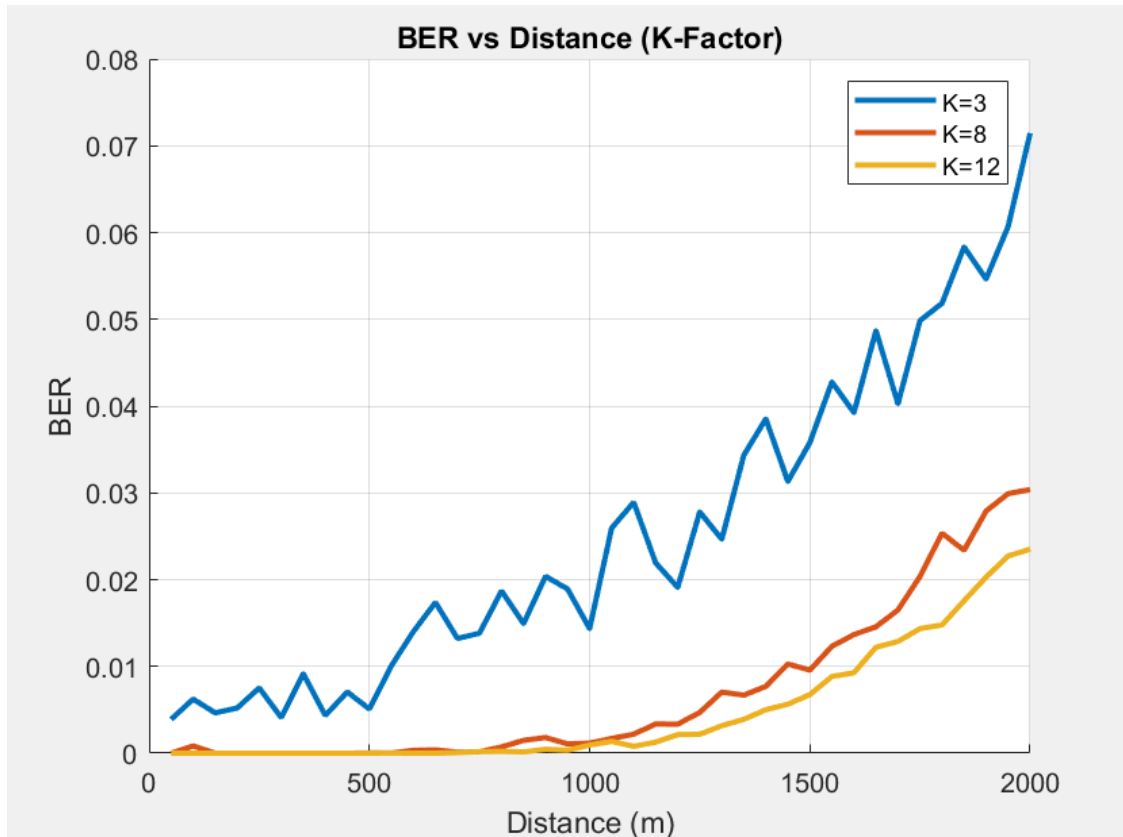


Figure 5: BER vs Distance (K-Factor)

Figure 5 demonstrates the effect of the Rician K-factor on system performance. A higher K-factor ($K = 12$), representing a strong Line-of-Sight (LoS) component, results in significantly lower BER across all distances. In contrast, for lower K-factor values ($K = 3$), where multipath fading is dominant, the BER increases rapidly with distance. This indicates that adaptive modulation must operate more conservatively in non-LoS environments.

Impact of Velocity (Doppler Effect):

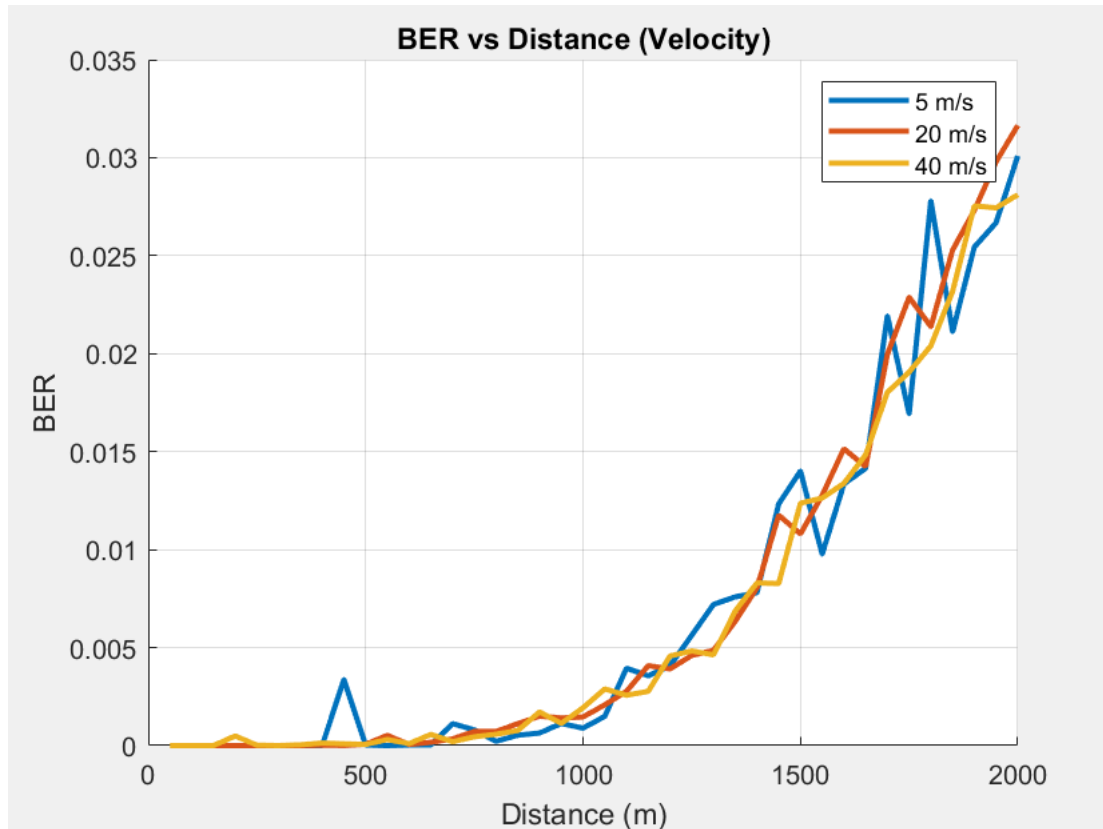


Figure 6: BER vs Distance (Velocity)

Figure 6 evaluates the effect of user mobility. As velocity increases from 5 m/s to 40 m/s, Doppler shift causes rapid channel variations, leading to increased BER. At higher velocities, the BER rises more sharply with distance, demonstrating that mobility degrades system performance. This highlights the importance of fast and accurate channel estimation in adaptive modulation systems.

Overall Observations

From the results, it is evident that the proposed adaptive modulation system effectively responds to variations in channel conditions. The system maintains reliable communication at low SNR by switching to robust modulation schemes, while exploiting favorable conditions to achieve higher data rates using higher-order modulation.

The combined analysis of modulation switching and BER performance confirms that adaptive modulation significantly improves system efficiency and reliability compared to fixed modulation schemes in dynamic wireless environments.

Discussion

The results obtained from the simulation highlight the effectiveness of the proposed adaptive modulation scheme in dynamically responding to varying channel conditions. This section discusses the broader implications of the observed results and provides deeper insights into system behavior.

Trade-off Between Reliability and Spectral Efficiency

One of the key observations from the results is the inherent trade-off between reliability and spectral efficiency. Lower-order modulation schemes such as BPSK provide high robustness against noise and fading but offer low data rates. In contrast, higher-order schemes like 16-QAM achieve higher spectral efficiency but are more susceptible to errors.

The adaptive modulation strategy effectively balances this trade-off by selecting the most appropriate modulation scheme based on SNR. This ensures reliable communication under poor channel conditions while maximizing throughput when the channel is favorable.

Impact of Channel Conditions

The performance of the adaptive system is highly dependent on channel characteristics such as path loss, fading, and frequency. Environments with higher path loss exponent or higher carrier frequencies result in rapid signal degradation, forcing the system to switch to more robust modulation schemes.

Similarly, the presence of a strong Line-of-Sight component (high K-factor) significantly improves system performance by reducing fading effects. Conversely, in environments dominated by multipath fading, the system must operate more conservatively to maintain acceptable BER levels.

Effect of Mobility

User mobility introduces Doppler shifts, causing rapid variations in the channel. As observed in the results, higher velocities lead to increased BER due to the difficulty in tracking fast-changing channel conditions.

This highlights the importance of efficient channel estimation and fast feedback mechanisms in practical adaptive modulation systems, especially in high-mobility scenarios such as vehicular communication.

Advantages of the Proposed Approach

The proposed adaptive modulation system demonstrates several advantages:

- Efficient utilization of available bandwidth
- Improved reliability under varying channel conditions
- Flexibility to adapt to environmental and system parameters
- Practical applicability in modern wireless systems

These advantages make adaptive modulation a suitable choice for next-generation communication systems.

Limitations of the Study

Despite its effectiveness, the proposed system has certain limitations. The model assumes perfect channel estimation and instantaneous feedback, which may not be achievable in practical systems. Additionally, the system does not consider channel coding or interference from other users.

Furthermore, the analysis is limited to a 2D channel model and three modulation schemes. Real-world systems may require more complex models and a wider range of modulation and coding techniques.

Conclusion

This paper presented an adaptive modulation switching system for a 2D wireless communication channel, designed to dynamically adjust modulation schemes based on varying channel conditions. The system utilizes Signal-to-Noise Ratio (SNR) as the key parameter for selecting between BPSK, QPSK, and 16-QAM modulation techniques.

The analysis presented in this study confirms that Adaptive Modulation (AM) is an essential framework for optimizing 2D terrestrial wireless communication. Through MATLAB simulations, it is demonstrated that fixed modulation schemes are fundamentally incapable of handling the dynamic nature of real-world channels, often leading to either link instability or inefficient utilization of available bandwidth.

Key findings from the research include:

- **Operational Efficiency:** The SNR-based switching logic successfully maintains link connectivity across a wide range of distances by downshifting to robust modulation schemes such as BPSK under poor channel conditions and upshifting to higher-order schemes such as 16-QAM when the channel is favorable.
- **Parameter Sensitivity:** System performance is significantly influenced by environmental and system parameters. Higher path loss exponents and higher carrier frequencies result in rapid signal degradation, reducing the effective range of high-order modulation schemes.
- **Reliability vs. Mobility:** The inclusion of Rician fading and Doppler effects demonstrates that while a strong Line-of-Sight component (high K-factor) improves reliability, increased receiver velocity introduces rapid channel variations that degrade performance and require adaptive strategies to maintain acceptable Bit Error Rate (BER).

The simulation results further validate that the proposed adaptive modulation strategy effectively balances reliability and spectral efficiency. The system ensures stable communication at low SNR while maximizing data throughput under

favorable conditions, thereby outperforming fixed modulation schemes in dynamic wireless environments.

Although the proposed model provides valuable insights, it is based on simplifying assumptions such as ideal channel estimation and absence of interference. Future work can focus on incorporating more realistic channel models, including shadowing effects, multi-user interference, and advanced techniques such as MIMO and machine learning-based adaptive modulation.

In conclusion, this study demonstrates that intelligently adapting modulation schemes in response to channel variations is crucial for achieving the optimal balance between data rate and reliability, making adaptive modulation a key enabler for modern wireless communication systems.

Future Work

Future work can focus on extending the proposed adaptive modulation system by incorporating more realistic channel models such as Rayleigh fading and shadowing effects. The inclusion of channel coding techniques can further enhance system reliability and reduce Bit Error Rate (BER).

Additionally, machine learning-based approaches can be explored for intelligent modulation selection under complex and rapidly varying channel conditions. The model can also be extended to three-dimensional communication environments, such as UAV-based systems, to analyze adaptive modulation in more dynamic scenarios.